

Pablo Alejandro López Sánchez

Data Engineer · AI & Automation

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PROFILE

Data Engineer specialized in automation and AI systems. I design and implement data architectures, MLOps infrastructure, and automation pipelines, and bring AI to the core of operations through agentic systems and models that turn manual processes into autonomous workflows. I build technical projects end-to-end, from data extraction through modeling, orchestration, and interface, integrating LLMs to automate analysis, generate insights, and support business decisions at scale.

SKILLS

Data Engineering: Python (Polars, SQLAlchemy, FastAPI), SQL (SQL Server, PostgreSQL), ETL Pipelines, Web Scraping (Selenium, Scrapy).

AI & Agentic Systems: Multi-agent development, spec-driven development, harness engineering, MCP (Model Context Protocol), Claude Code / Agent SDK, prompt engineering.

Infrastructure & Orchestration: Dagster, Docker, SSH/SFTP, Linux Server, Git/GitHub, Data Flow Modeling.

Analytics: Power BI (DAX, Power Query), KPI Analysis.

Business Strategy: Project Management, Demand Planning, Inventory Optimization.

Languages: English (B2-C1), Spanish (Native).

EXPERIENCE

Data Engineer – Mercaldas, Manizales, Colombia

July 2025 – Present

Leading the company's data and automation engineering, driving end-to-end the shift from siloed, manual processes to a centralized architecture with in-house MLOps, autonomous demand forecasting, and price intelligence, plus internal tools already adopted across multiple business areas.

- **MLOps Infrastructure & Data Autonomy:** Led the design and implementation of an end-to-end MLOps infrastructure for autonomous demand forecasting, reaching **70-82%** forecast accuracy depending on the cluster and driving a decisive reduction in **stockouts** from a historical **12-20%** range to around **8%**.
- **Market Price Intelligence:** Designed and built an internal analytics platform that forecasts wholesale agricultural product prices **52 weeks** ahead using per-cluster ML models (**XGBoost/LightGBM**) and generates **automated AI insights** (Claude), delivered through a custom web app that benchmarks the market against internal prices to anticipate purchasing decisions.
- **Stock-Transfer Planning Automation:** Built the web tool the **purchasing analysts** use to plan inventory transfers between stores; it cuts each transfer plan from hours to minutes (**1-4 hours saved per run**) and improves merchandise-reallocation decisions, reducing overstock and stockouts across stores.
- **Centralized Orchestration (Dagster):** Deployed and configured an orchestration platform on a dedicated **Linux** server to unify dispersed processes. Designed to automate manual executions and optimize failure monitoring, resulting in an estimated reduction of ~4 operational hours per week per user for the IT and Demand Planning teams.
- **Governance & DevSecOps:** Established the corporate development ecosystem on **GitHub**, implementing coding rules, version control, and **agentic workflows** using Claude AI to standardize internal software production.

Business Analyst – Latin America Sourcing, Remote

May 2025 – July 2025

- Managed and analyzed commercial data using Airtable and Stripe to optimize negotiation processes with international suppliers.
- Analyzed client requests, documented commercial interactions, and generated operational insights. Automated supplier classification processes using custom Python scripts.

Business Intelligence Internship – Industria Licorera de Caldas, Manizales, Colombia

Feb 2023 – Jul 2023

- Developed Power BI dashboards for Operations and Foreign Trade departments, integrating data from Dynamics 365 to drive logistics efficiency.

PROJECTS

Five production systems, each with an interactive walkthrough: pablo.maieutik-data.com.

ML demand forecasting: Weekly forecasts for 100,000+ product and store combinations across 14 stores, feeding replenishment. 19% lower error than the system it replaced.

Market price intelligence: Ingests Colombia's weekly wholesale price bulletins, forecasts 52 weeks per product, and has a model write the reading of each series.

AI assistant for business data: Telegram assistant answering business questions in plain language, with every figure traced back to a real query against the warehouse.

Internal operations platform: Central API for recurring operations work. Transfer planning went from roughly 4 hours to 2-5 minutes, run by the team itself.

Personal AI operating system: Personal project. A Git-versioned Markdown vault as long-term memory, with a multi-agent runtime that asks before anything it cannot undo.

EDUCATION

Bachelor of Business Administration (BBA) – Universidad Nacional de Colombia, October 2023

CERTIFICATES & COURSES

Full certificates list: [here](#)

Google Data Analytics Professional Certificate – Google | Coursera

Databases for Data Scientists Specialization – University of Colorado Boulder | Coursera

Claude Code in Action – Anthropic Education

Data Analysis Tools – Wesleyan University | Coursera

REFERENCES

Luisa Soraya Santana Murcia – Mercaldas | Demand Planning Leader, +57 318 358 9892, ssantana@mercaldas.com.co

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